Example - Concrete Slab

```
close all;   clear all
%Define vertex coordinates
Vertices=[0, 0;   8, 0;  8, 5;  0, 5];

%Define segments
Segments=[  1, 2; 2, 3; 3, 4; 4, 1];

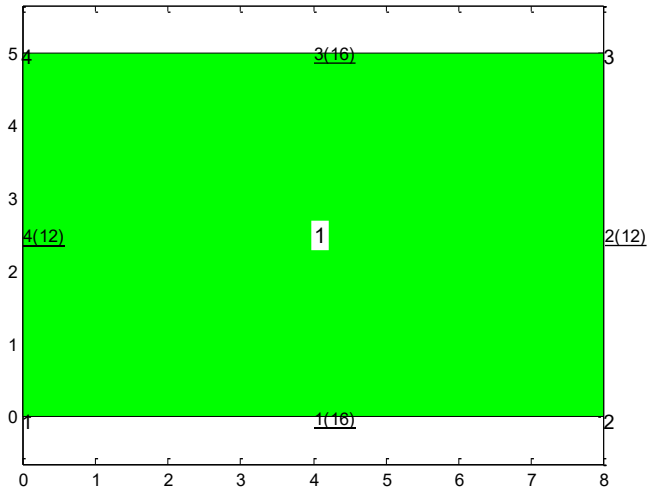
%Define surfaces
Surfaces=[1 2 3 4];

%Define number of elements per segments
Seed=[8 6 8 6 ]*2;

%Define number of element nodes and dofs per node
nen=4;
dofsPerNode=3;
Segp=[Seed];
mp=[dofsPerNode, nen];

%Draw geometry
geomdraw2(Vertices,Segments,Surfaces,Segp,mp)

%Generate element mesh
[Coord Edof Dof meshdb]=strMeshgen(Vertices,Segments,Surfaces,Segp,mp);
```



Example - Concrete Slab

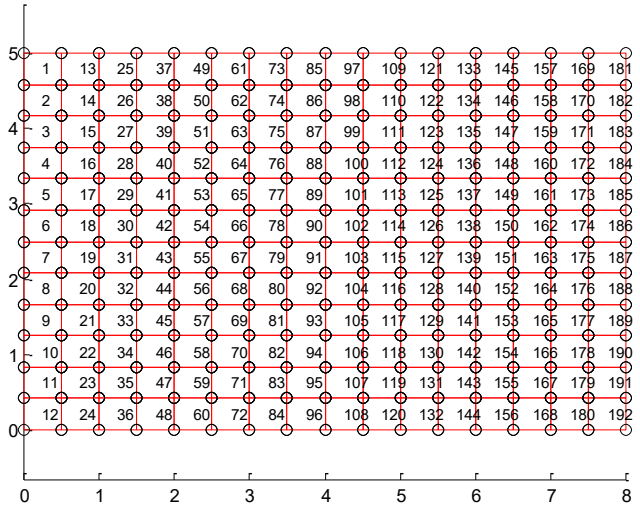
```
%Generate element coordinates
[Ex,Ey]=coordxtr(Edof,Coord,Dof,nen);

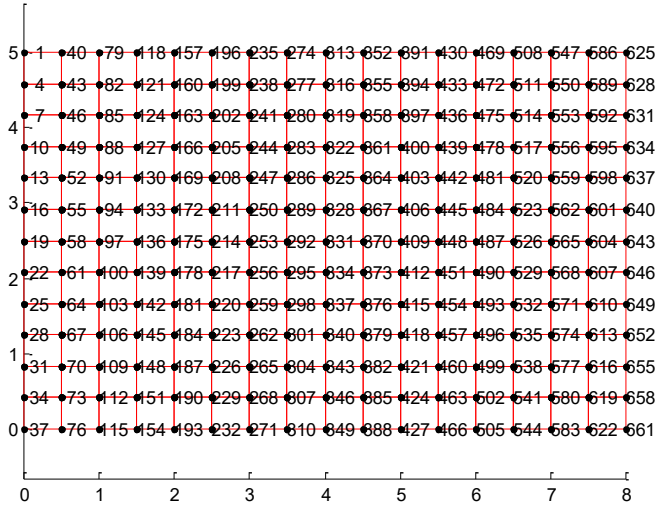
%Draw element mesh with numbering of the elements
eldraw2(Ex,Ey,[1 4 1],Edof(:,1));

%Draw element mesh with numbering of the 1 in each node
eldraw2(Ex,Ey,[1 4 0]);
text(Coord(:,1),Coord(:,2),num2str(Dof(:,1)))

%Prescribed displacements
bc1=extrSeg([1 2 3],meshdb,1);
bc2=extrSeg([1],meshdb,2);
bc=[bc1, 0*ones(length(bc1),1); bc2, 0*ones(length(bc2),1)];

%Element and material properties and load
ep=[0.15];
eq=[0.91*(-4.9e3*1.2-2e3*1.5)];
ptype=1;
E=30e9;
v=0.2;
D=hooke(ptype,E,v);
nel=size(Edof,1);
ndof=max(Edof(:));
%
K=sparse(ndof,ndof);
f=sparse(ndof,1);
```





Example - Concrete Slab

```
%Calculating element stiffness matrices and assembling to global stiffness matrix
for i=1:nel
   [Ke,fe]=platre(Ex(i,:),Ey(i,:),ep,D,eq);
   [K,f]=sparse_assem(Edof(i,:),K,Ke,f,fe);
end

%Solving equation system
[a,r]=solveq(K,f,bc);
Ed=extract_ed(Edof,a);

%Calculating element moment and shear forces and curvature
Es=[]; Et=[];
for j=1:nel
   [Es(j,:),Et(j,:)]=platrs(Ex(j,:),Ey(j,:),ep,D,Ed(j,:));
end

%Draw the displacements
figure(4)
fill3(Ex',Ey',Ed(:,1:3:12)',Ed(:,1:3:12)')
grid on
```

